

Postman & API Monitoring: Logs & Tester Mindset

Monitoring and logging are essential for **understanding API behavior, debugging issues, and ensuring traceability**. This guide provides a tester-focused approach to logs, correlation IDs, and prioritization.

1. Key Concepts

App Logs

- Generated by the **application itself** (backend service).
- Includes:
 - Business logic execution
 - Errors and exceptions
 - Warnings, info messages, and debug traces
- Purpose: Identify **why an API or feature failed internally**

API Gateway / Proxy Logs

- Generated at the **gateway or API layer** (e.g., Nginx, AWS API Gateway, Kong).
- Includes:
 - Request metadata (method, headers, URL)
 - Response codes and latency
 - Authentication/authorization events

- Purpose: Identify **network, routing, or policy-level issues**
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2. Correlation IDs

- A **unique identifier** assigned to a request to track it across services and layers.
- Useful when debugging issues in distributed systems.

How to use:

1. Pre-request: generate or capture correlation ID
2. Include in headers (**X-Correlation-ID**)
3. Use the same ID to trace through:
 - API Gateway logs
 - App logs
 - Downstream services

Example Header:

X-Correlation-ID: CID-1675630200

3. Which Logs to Check First and Why

When an API fails or behaves unexpectedly, follow this **priority sequence**:

Step	Log Type	Reason
1	API Gateway / Proxy Logs	Check if the request reached the server , status codes, latency, authentication, or routing issues. Fastest way to identify network or gateway-level failures.

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|---|-------------------------------------|---|
| 2 | Application Logs | Once the request reaches the app, check business logic execution , error traces, and exceptions. Correlation IDs help pinpoint the exact request flow. |
| 3 | Downstream Service / DB Logs | If the API depends on other services or databases, check these logs to see if external dependencies caused the failure. |

Why this order:

- API Gateway logs are usually **lighter, quicker to access**, and reveal **request reachability**.
 - App logs are **heavier and more detailed**, best examined after confirming the request reached the server.
 - Correlation IDs ensure **you're tracking the same request across all logs**, avoiding wasted debugging time.
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4. Tester Mindset for Logs

- Always **start from the top-level log (gateway/proxy)** → verify request arrival
 - Trace with **correlation IDs** → follow the request end-to-end
 - Identify **if the failure is client-side, network, app logic, or downstream dependency**
 - Document **timestamps, request IDs, and errors** for reproducibility
 - Ask: *"If I fix the code, will this log tell me the issue is gone?"*
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5. Best Practices

- Ensure all requests **include correlation IDs** during testing
- Regularly review **API gateway logs** for anomalies even if no failures are reported

- Combine logs with **Postman console and test scripts** to create an end-to-end picture
- Maintain a **log checklist**: Gateway → App → Downstream → Metrics